

RECONSIDERING THE STRUCTURE OF PIPELINED 8051 MICROPROCESSOR

Mr. Narongrat Srimee

Mr. Nattapong Thongprom

Asso.Prof. Somsak Mitatha Advisor

Mr. Chae n Vongchumyen Advisor

Academic Year 2004

ABSTRACT

This project is the design of pipelined 8051 microprocessor use VHDL language to describe the function of circuit. In the design process, the researcher use top-down design model to develop the microcontroller by separate it to stage and separate each stage to component. Then write each component in VHDL language and test it with simulation software to ensure that the component is working correctly. After that, combine it to their stage and test them together for the corrective of cooperation between these components. Later, combine each stage again to the microcontroller and test it by simulation software. Then bring the microcontroller derived soft-core into synthesis process to convert VHDL code to gate-level netlist and bring this netlist into place-and-route process to program it to FPGA. Finally, test the programmed FPGA on circuit board to test the corrective operation of it.

The designed microcontroller improves the speed of operating in term of Clock Per Instruction (CPI). The capability of 62 instructions from based microcontroller. The maximum frequency of this microcontroller can run is 25 MHz and working correctly.